EFFECT OF FREQUENCY-TO-PLACE MISMATCH AND FREQUENCY WARP ON SPEECH AND MUSIC SOUND QUALITY IN ACOUSTIC COCHLEAR IMPLANT SIMULATION

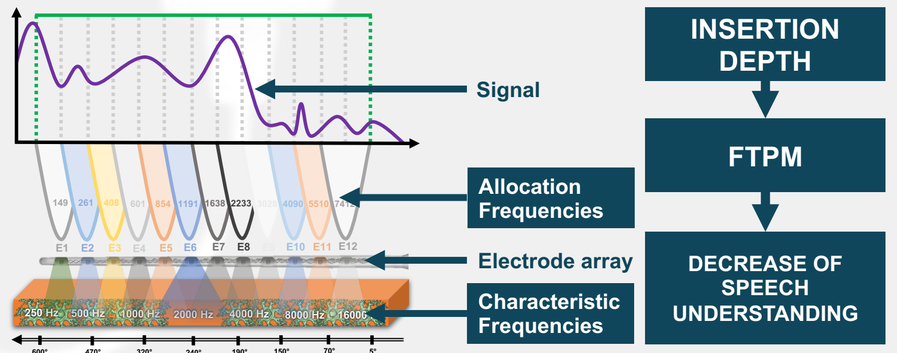
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In more recent years a greater emphasis has been placed on improving the **sound quality** for individuals using cochlear implants (CIs). Indeed, many post-lingually deafened adults with CIs have speech recognition performance close to normal hearing (NH) listeners in quiet environments, but yet complain about poor sound quality, especially for music. The contribution of CI technology to the **degradation of music quality** has been well documented, but little is known about the relative importance of different aspects relating to the technology and how they interact with the personal preferences and hearing characteristics of the individual.

One individual factor that is suspected to contribute largely to the sound quality degradation is the frequency-to-place mismatch (FTPM) induced by the incomplete insertion of the electrode array inside the cochlea.

FTPM is defined as a deviation between the characteristic frequency and the allocation frequency assigned during CI fitting.



WHAT IS THE IMPACT OF FREQUENCY-TO-PLACE ON SOUND QUALITY?



RATIONALE

Whereas much of the speech perception improvement occurs during the first months after CI activation, recent studies have shown that a significant FTPM could negatively affect this adaptation process. Alongside the improvements in speech perception over time from device activation patients often also change their reports of sound quality, from "high pitched", "screechy", "dalek-like" to more "normal" or "natural" after extensive listening experience.

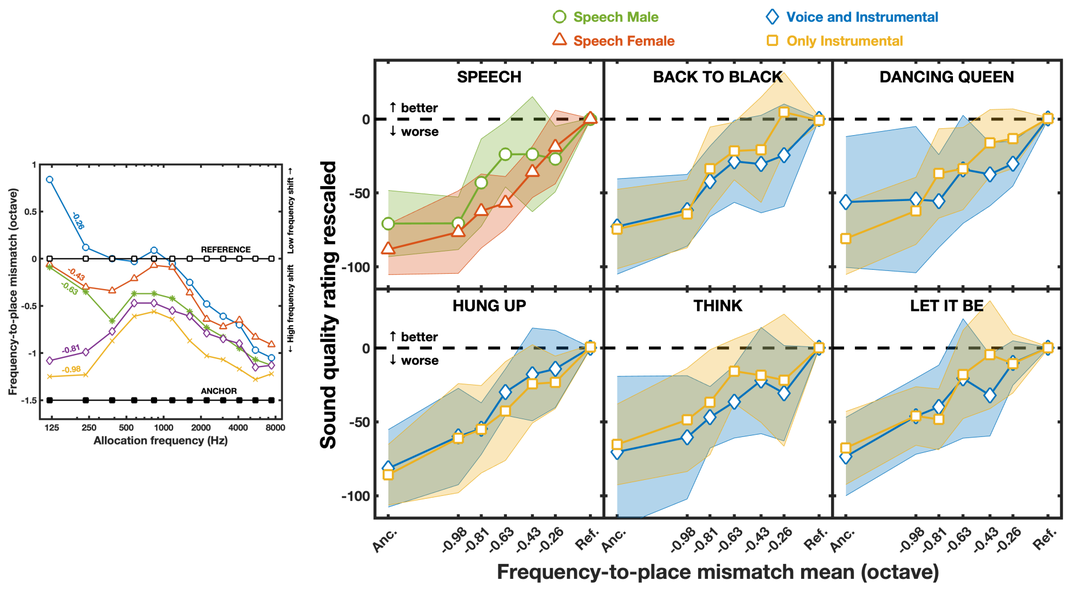
However, the specific contribution of FTPM on sound quality isn't clear yet.

FTPM PROFILES → NOISE VOCODER → MUSHRA TEST → CORRELATION FTPM / SOUND QUALITY

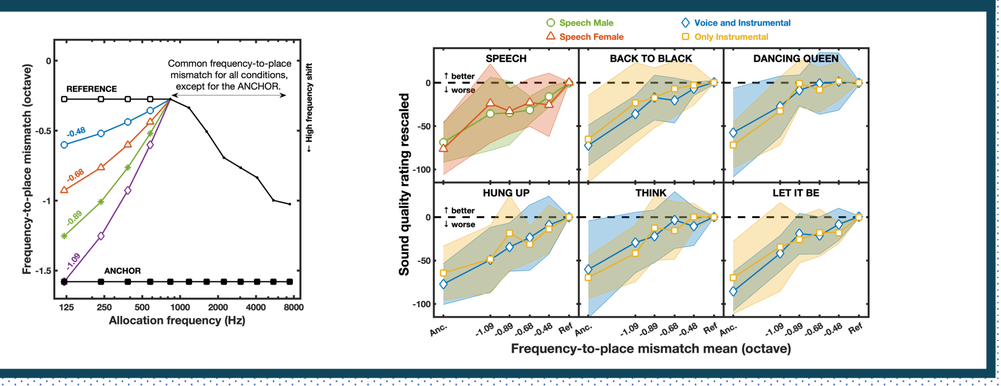
METHOD

- 8 NH listeners tested
- FTPM from Canfarotta et al. (2022)[1]
- Simulated using a **noise vocoder**[2]
- The sound quality was evaluated for the different FTPM profiles using **MUSHRA test**
- **Different sound materials** were tested: speech, music original, music instrumental

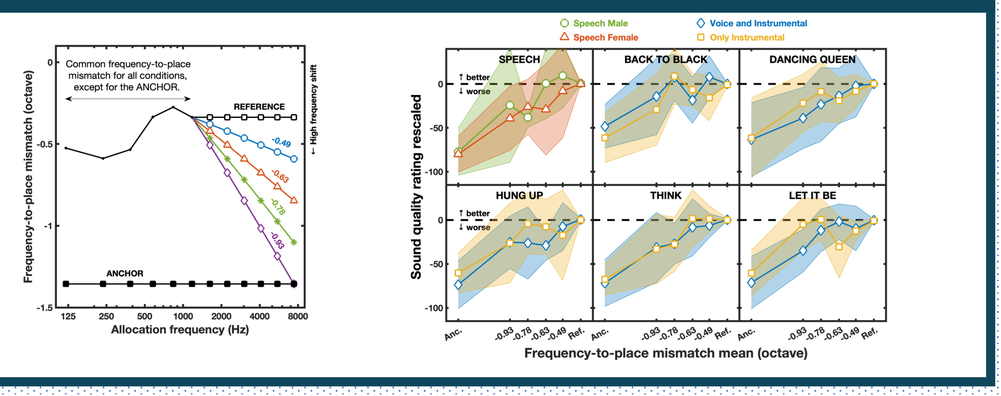
EXPERIENCE 1 – EFFECT OF REALISTIC FTPM



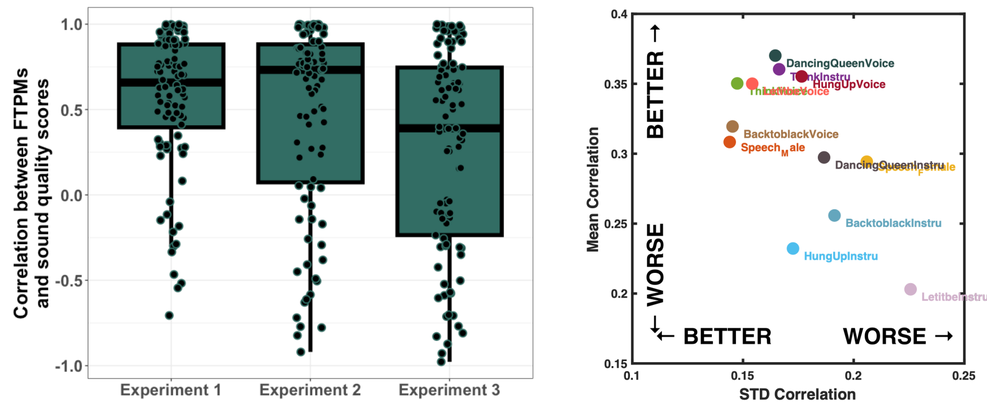
EXPERIENCE 2 – EFFECT OF LOW FTPM



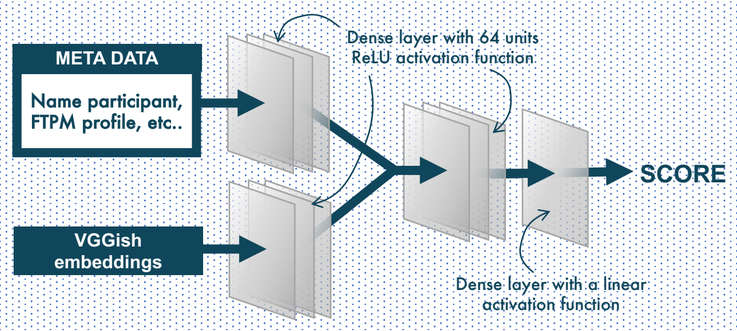
EXPERIENCE 3 – EFFECT OF HIGH FTPM



SUMMARY

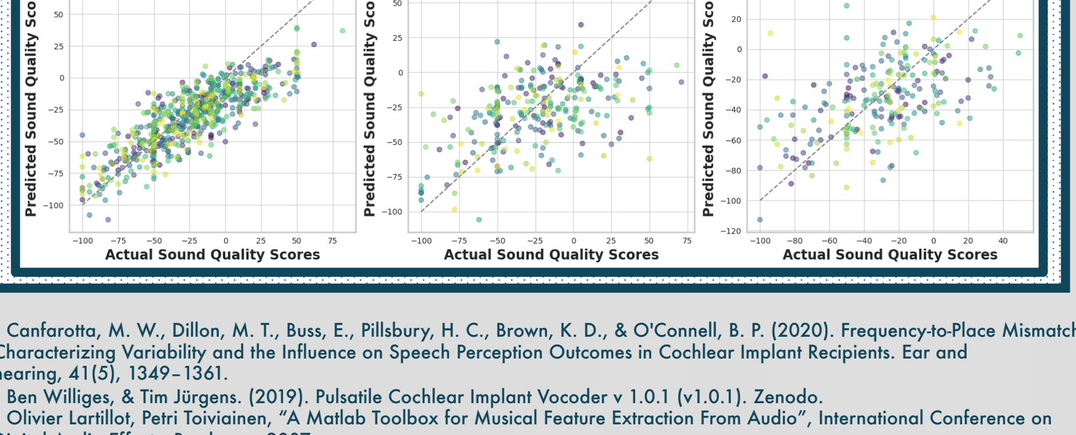


HOW EXPLAIN THESE DIFFERENCES ?



TEST LOSS 759.43

MEAN ABSOLUTE ERROR 21.44



CAN WE PREDICT FREQUENCY-TO-PLACE MISMATCH USING FREQUENCY SHIFT?

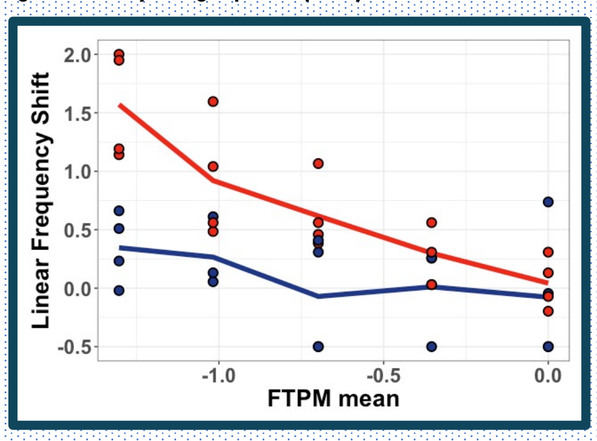
RATIONALE

SPEECH & MUSIC → DIFFERENT LINEAR FREQUENCY SHIFT → FTPM profile → NOISE VOCODER → MUSHRA TEST → EXTRACTION OF THE LINEAR FREQUENCY SHIFT WITH THE BEST SOUND QUALITY

FTPM profile → FTPM mean → CORRELATION *(see results)*

It is possible to investigate electrode placement (and FTPM) with imaging measures, such as multisection computed tomography (CT) scans, x-rays or conebeam CT. However, these measures are costly, their quality is not always precise enough to see cochlea structures or are just not always available clinically. Consequently, characterization of FTPM in clinics is rare. Without that information, the acclimatization process cannot be fully anticipated and there are no opportunities to intervene to maximise the adaptation process, if required.

Here we aim to investigate FTPM by using a pre-frequency-shifted sound.



[1] Canfarotta, M. W., Dillon, M. T., Buss, E., Pillsbury, H. C., Brown, K. D., & O'Connell, B. P. (2020). Frequency-to-Place Mismatch: Characterizing Variability and the Influence on Speech Perception Outcomes in Cochlear Implant Recipients. Ear and hearing, 41(5), 1349–1361.
[2] Ben Williges, & Tim Jürgens. (2019). Pulsatile Cochlear Implant Vocoder v 1.0.1 (v1.0.1). Zenodo.
[3] Olivier Lartillot, Petri Toiviainen, "A Matlab Toolbox for Musical Feature Extraction From Audio", International Conference on Digital Audio Effects, Bordeaux, 2007.

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